MONASH University
Information Technology

FIT3080 – Artificial Intelligence

Introduction to Artificial Intelligence Chapter 1

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A Brief History of AI and State of the Art

"Those who don't remember history are condemned to repeat it" – George Santayana

Dartmouth Conference (1956)

Goals of the Dartmouth Conference

*We propose that a **2-month, 10-man study of artificial intelligence** be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that **every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it**. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. **We think that a significant advance can be made** in one or more of these problems if a carefully selected group of scientists work on it together for a summer.*

History of AI (I)

- 1943 Perceptrons/Neural nets/Connectionism (McCulloch and Pitts 1943, Rosenblatt 1957)
- 1950s Machine translation
- 1950 Turing initiated AI as a research area
- 1956 Dartmouth conference: Birth of AI
  - Origin of *Artificial Intelligence* as a name
- 1963 Checkers playing (Samuel 1963)
- 1963 Theorem Prover (Newell 1963)
  - GPS - General Problem Solver (Newell, Shaw & Simon)
    Basic technique: Means-ends analysis
- 1964 Bayesian inference applied to authorship attribution (Mosteller and Wallace 1964)
- 1965 Robinson's complete algorithm for logical reasoning

History of AI (II)

- 1974 Neural networks research almost disappears
- **1974 Winter of AI:** no world knowledge and no scaling up
- 1969-79 Knowledge-based systems (MAXIMA, MYCIN)
- 1980-88 AI becomes an industry: Expert systems, robots
- 1986 Neural networks return to popularity
- 1987 Return of probability; increase in technical depth
  - ALife, Genetic Algorithms, soft computing
- **1988 Winter of AI:** cancellation of new spending on AI
- 1995 Intelligent agents
- 2001 Big data, Deep learning
- 2011 Turing Award (Pearl) for Probabilistic AI reasoning
- 2018 Turing Award (Bengio, Hinton, LeCun): Deep NNs
- **Is another AI winter coming?**

State of the Art (I)

- Autonomous agents – Smart spaces/ambient intelligence
- Recommender systems – Guides (travel, museum, shopping), directed advertising
- Machine learning applications – e.g., spam fighting, disease diagnosis, data mining (business intelligence)
- Google's search engine (1998) – Page rank algorithm
- Autonomous planning and scheduling (1991, 1999, 2004, 2008)
- Robotic vehicles – autonomous driving (1995, 2006, 2007, now) – (https://www.google.com/selfdrivingcar/)
- Robotics – Roomba (2002), packBot (2002)

State of the Art (II)

- Game playing –
  - Deep Blue defeated the world chess champion Garry Kasparov (1997)
  - AlphaGo defeated the world Go champion Lee Sedol (2016)
- Statistical machine translation (2007) – (https://translate.google.com)
- Winning Jeopardy – Watson (2011)
- Spoken conversational agents – e.g., Siri (2011), Alexa (2014)
- Generative AI technologies – e.g., ChatGPT, Claude (2020s)

Which of the following can be done now?

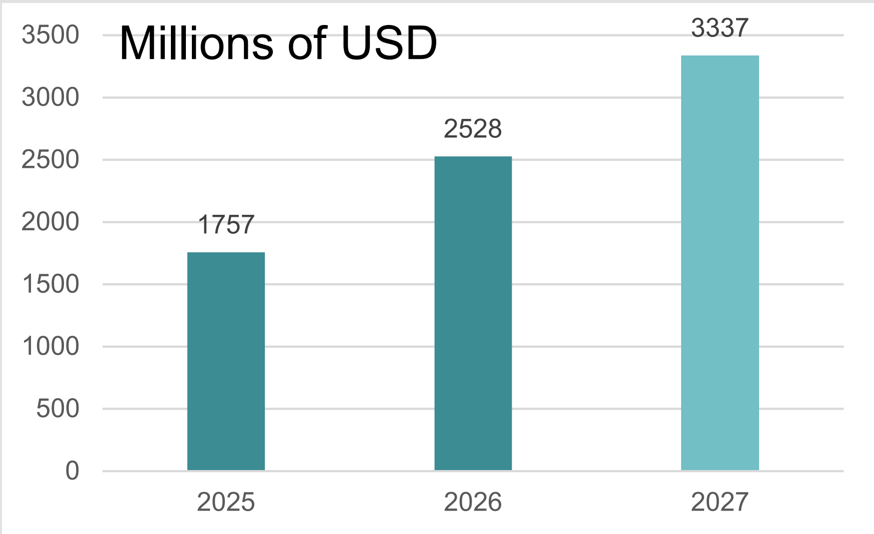
- Play decent table tennis
- Drive a curving mountain road
- Drive in Brunswick St on Saturday night
- Play decent bridge
- Prove a new maths theorem
- Write a funny story
- Provide competent specialist legal advice
- Real-time English → Japanese speech

AI has become mainstream

- Industrial Revolution → major increase in manufacturing productivity
- AI Revolution → promises major increase in knowledge work productivity

AI is receiving significant attention and investment

https://www.gartner.com/en/newsroom/press-releases/2026-1-15-gartner-says-worldwide-ai-spending-will-total-2-point-5-trillion-dollars-in-2026#



But there can always be another winter …

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Intelligence and Artificial Intelligence

What is Intelligence?

An entity is intelligent if

- It has internal knowledge
- It has world knowledge
- It has intentions and plans to fulfill these intentions
- It can communicate
- It has creativity

What is Artificial Intelligence (AI)?

- ***AI is the study of mental faculties through the use of computational models***

  Charniak and McDermott, 1985

- ***AI is the study of how to make computers do things that (at the moment) humans do (better)***

  Rich and Knight, 1991

- ***AI is the science of making computers act like the ones in the movies***

  Anonymous

AI is 20 years away

Goals of AI Research

- **Find out about the nature of intelligence**
- **Build an intelligent machine**

Build systems that	
Think like humans	Think rationally
Act like humans	Act rationally

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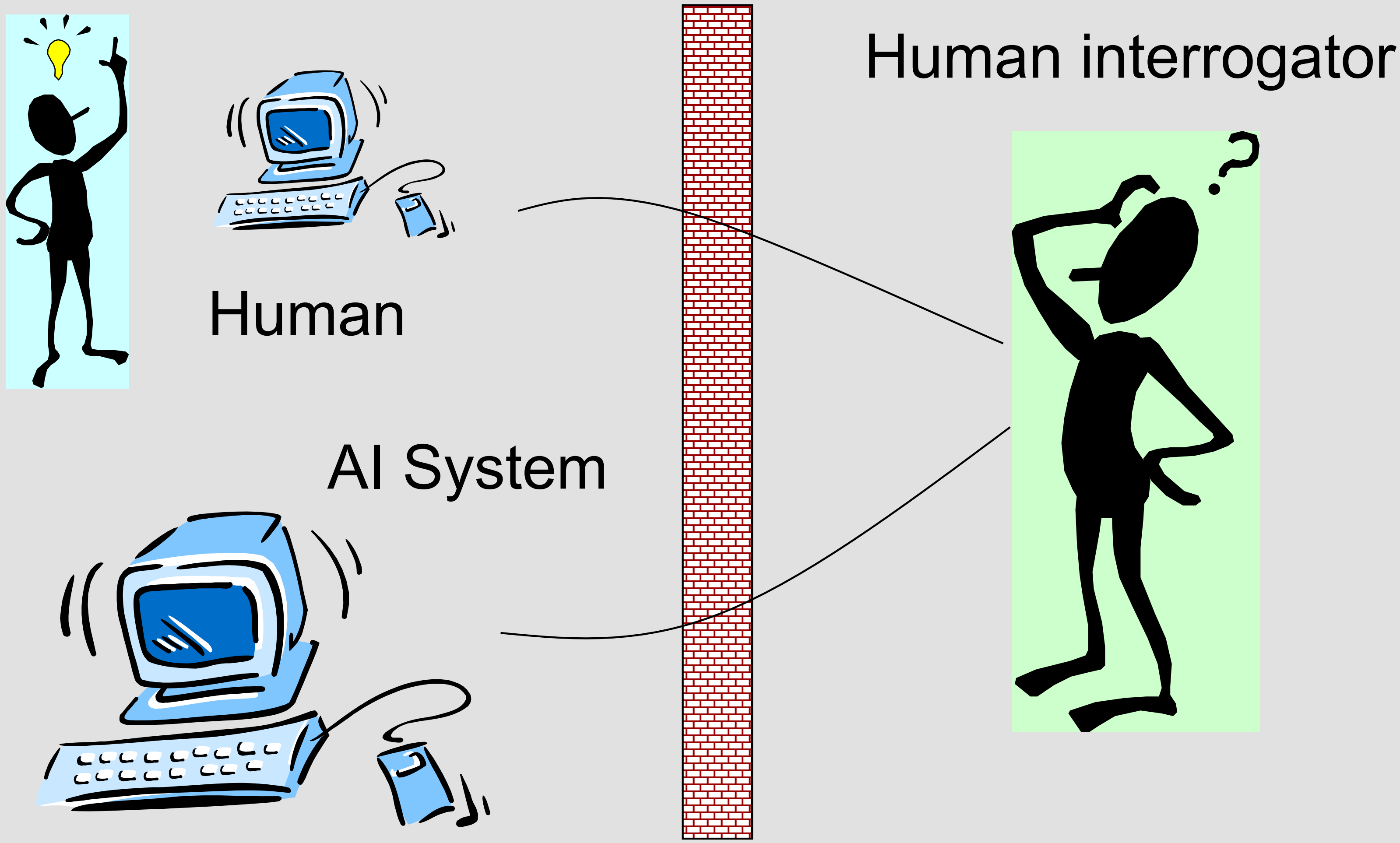
The Turing Test

Acting Humanly – The Turing Test (I)

Turing test (1950)



Acting Humanly – The Turing Test (II)

Turing (1950)

- https://plato.stanford.edu/entries/turing-test/
- **Can machines think? → Can machines behave intelligently?**
  - Operational test for intelligent behaviour

☺ **Suggested major components of AI: knowledge, reasoning, language understanding, learning**

☹ **Not reproducible, constructive, or amenable to mathematical analysis**

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Rationality

Acting Rationally and Autonomous Agents

- **Rational behaviour: doing the right thing**
  - The right thing: that which is expected to maximize goal achievement, <u>given the available information</u>
- **Agency**
  - Having internal goal structure and external behaviour which generally serves to satisfy a goal structure
- **Agent**
  - An agent is an entity that perceives and acts
- **Autonomy**
  - Ability to operate independently

Our objective: Build autonomous, rational agents

Rational Agents and Bounded Rationality

- Abstractly, an agent is a function from percept histories to actions:

$$f: \mathcal{P}^{*} \rightarrow \mathcal{A}$$

- For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance
- Caveat: computational limitations make *perfect rationality* unachievable → *bounded rationality*
  - design the best program for a given machine's resources

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Sub-fields of AI

Sub-fields of AI

- **Search**
- **Knowledge representation and reasoning**
  - **Logic-based**
  - **Probabilistic**
- Planning and decision making
  - **MDPs**
- Multi-agent systems
- **Machine learning**
  - **Supervised**
    - **Classification**
    - **Regression**
  - **Unsupervised**
    - **Clustering**
    - Anomaly detection
  - **Reinforcement**
  - **ANNs** / Deep

Strong versus Weak AI

‘Strong’ AI	‘Weak’ AI
Build a machine that is **actually thinking**	Build a machine that **acts as if it were intelligent**
Top-down perspective: does my program **think**?	Bottom-up perspective: does my program **work**?
Create **general strategy** to intelligence: • One program that can do any task	Create **domain-specific strategies**: • A chess playing program • A language translation program

Scope of FIT3080

This unit introduces the main problems and approaches to <u>designing AI systems</u> including

- Abstracting problems to facilitate solving them
- Finding the best action to perform (Search)
- Representing and reasoning about information (Logic)
- Reasoning under uncertainty (Bayesian networks)
- Determining the best sequence of actions to take (MDPs)
- Learning to perform well in unknown environments (RL)
- Generalizing to unseen data from historic examples (ML)
  - machine learning paradigms

Reading

- **Russell, S. and Norvig, P, *Artificial Intelligence – A Modern Approach* (4th ed), Prentice Hall, Chapter 1**
- **Other references**
  - W.S. McCulloch and W. Pitts (1943) A logical calculus of the ideas immanent in nervous activity. *Bull Math Biophysics*, 5, 115-137
  - Alan Turing (1950) Computing machinery and intelligence. *Mind*, 59, 433-460. Reprinted many times (e.g., M. Boden (ed) *Philosophy of AI*, Oxford, 1990)
  - F. Rosenblatt (1957) The Perceptron. Report 85-460-1 Cornell Aeronautical Lab
  - M. Minsky and S. Papert (1969) *Perceptrons*. MIT
  - A. Newell & H.A. Simon (1976) Computer science as empirical inquiry. *Communications of the ACM*, 19. Reprinted in Boden
  - M. Boden (1977), *Artificial Intelligence and Natural Man*. Basic Books Inc.

Next Lecture Topic

- **Lecture Topic 2**
  - Intelligent Agents